Sparks in detector 7, and how they compare to detector 3

(June cosmic bench — `g_det7_long`)

MX17 cosmic-bench QA

July 5, 2026

1 Purpose

The det3 spark study (companion note `det3_spark_analysis`) found sparks that are Poisson in time, predominantly muon-induced, edge-localised, and neither a single flash nor a propagating discharge. det7 is the most spark-prone detector of the June fleet (38.9 % of firing events spark, vs 9.1 % for det3) and runs at a different operating point (drift 700 V, resist past optimum). We repeat the identical analysis on `mx17_det6_det7_overnight_6-26-26 / long_run` (det7 = FEU6/8, 3.84 h, **22 247 sparks**) to ask: is it the *same* phenomenon, just more of it?

2 Answer: yes — the same story, amplified

Every det3 conclusion reproduces in det7, scaled up by the higher operating gain (Table 1). Sparks are muon-seeded avalanche breakdowns whose probability and spatial/temporal extent grow as the detector is pushed closer to breakdown, initiated at the same edge / hot-strip weak spots, developing as a spatially-extended, temporally-spread event with no propagating front.

	det3 (`g_det3_wknd`)	**det7** (`g_det7_long`)	reading
spark rate (of firing events)	9.1 %	38.9 %	det7 deep in sparking regime
mean rate	0.33 Hz	1.61 Hz	higher Poisson mean
random in time?	Poisson	Poisson	neither bursts (no HV instab.)
spark rate, muon *crosses* det	4.9 %	31.0 %	muon-induced
spark rate, muon *misses* (>30mm)	1.2 %	5.3 %	(stronger in det7)
crossing/miss enhancement	**4.1**×	**5.9**×	muon-dominated in both
spontaneous floor (miss rate)	1.2 %	5.3 %	floor rises with gain
sparks with no clean M3 track	60 %	47 %	DAQ cross-talk in both
strips/spark (X, Y)	40, 53	64, 76	bigger discharges; Y>X in both
spatial span (median)	∼200 mm (half)	∼360 mm (whole)	det7 fills the plane
occupancy peaks at	edges + hot strips	edges + hot strips	same weak spots
hit-time spread (median)	253 ns	334 ns	longer in det7
fraction $\sigma_t > 300$ ns	24 %	66 %	det7 rings longer
position–time corr $\lvert r \rvert$	0.12	0.15	no "walk" in either

Table 1: det3 vs det7 — identical phenomenology, det7 simply further into breakdown.

3 Q1 — Time: Poisson, faster

det7 sparks at a flat 1.61 Hz with no bursts; inter-spark intervals are exponential and counts-per-bin have std = $\sqrt{\text{mean}}$ (Fig. 1) — Poisson, exactly as det3, just a 5× higher mean. The elevated rate is therefore *not* a new instability; it is the same random process with more muons reaching breakdown.

(Mild consecutive-event clustering, $P(\text{spark} \mid \text{prev spark}) = 39\,\%$ vs $32\,\%$ baseline, matches det3's small excess.)

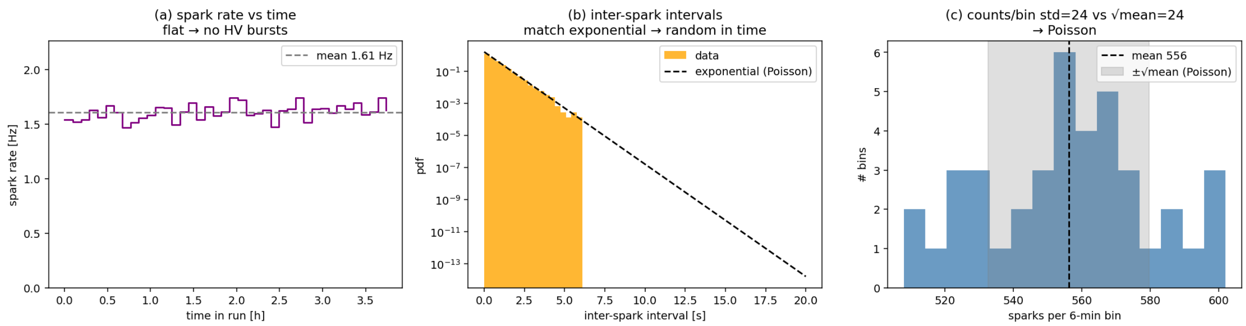

Figure 1: det7 time structure: flat rate (a), exponential intervals (b), Poisson counts/bin (c).

4 Q2 — Muon-induced, even more so

Conditioning on a clean single M3 muon, a track crossing det7 sparks at **31.0 %** versus **5.3 %** for one that clearly misses — a **5.9**× enhancement (Fig. 2a), stronger than det3's 4.1×. So ∼83 % of in-detector sparks are muon-induced. The spontaneous floor is higher than det3 (5.3 % vs 1.2 %), consistent with running closer to breakdown where even a stray electron can trigger a discharge. The discharge centroid is again central and compact (Fig. 2c). The M3 crossing map (Fig. 2b) shows more structure than det3's — but this tracks the overall M3 track density / telescope acceptance, not a det7 hot-spot; the detector-frame centroid is the clean spatial measure. The "no clean track" population persists (47 % of sparks vs 35 % baseline; same 1.3× over-representation and same discharge size), the DAQ common-mode cross-talk flagged for det3.

Shower hypothesis ruled out (as in det3). If the no-track sparks were multi-particle showers they would leave M3 with ≥ 2 track candidates; instead only **0.5 %** of det7 no-track sparks do (they are 62 % M3-blank, 38 % single degraded track), and their discharge morphology is *identical* to muon-tracked sparks (median 122 vs 125 strips, span 389 vs 392 mm, spread 335 vs 333 ns; Fig. 3). A spark nearly doubles M3's no-track rate (18 → 29 % blank) — cross-talk, not showers. This reproduces the det3 result on a 2.5× larger spark sample.

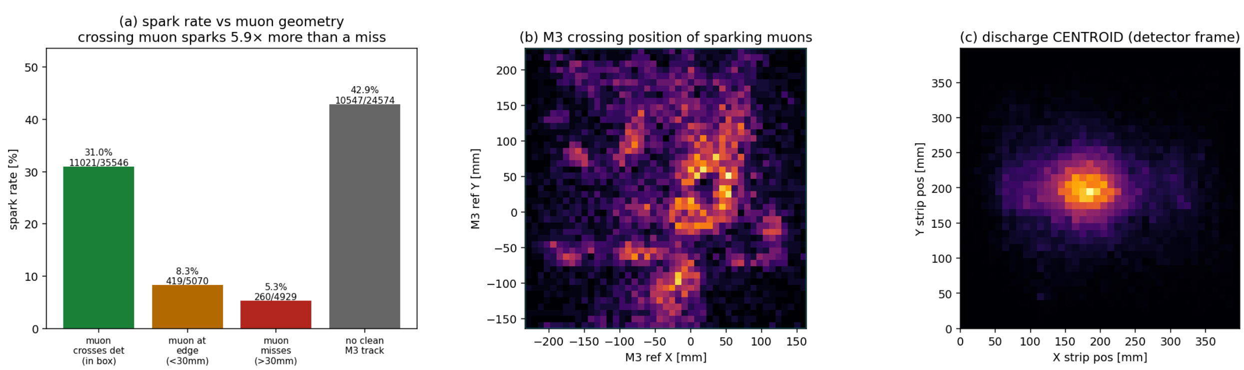

Figure 2: det7 muon dependence: (a) crossing muons spark 5.9× more than missing ones; (b) M3 crossing positions; (c) central discharge centroid.

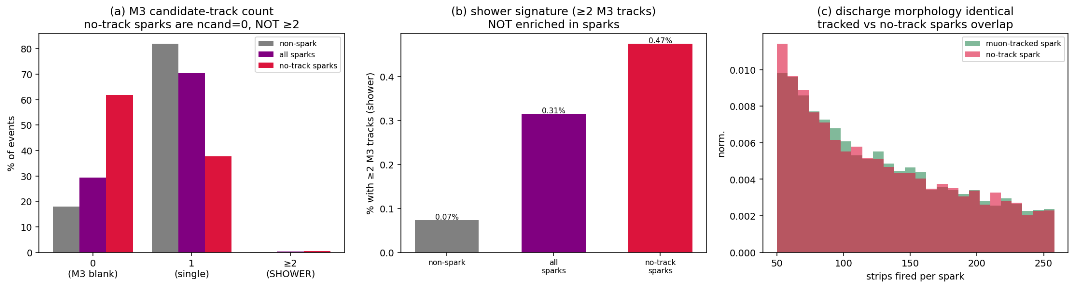

Figure 3: det7 shower test: (a) M3 candidate-track count by spark class; (b) the ≥ 2-track shower fraction stays at the $\sim$0.5 % level; (c) no-track and muon-tracked sparks have identical discharge size — showers ruled out.

5 Q3 — Same places: edges and hot strips

det7 strip occupancy peaks at the plane **edges** (both planes; strongest at low-X and high-Y) and on a few **hot strips** — most dramatically an X/FEU6 channel near 45 mm that fires in a large fraction of all sparks (Fig. 4a), plus fixed Y/FEU8 spikes (Fig. 4b). This is the same edge+hot-strip signature as det3, more pronounced. Discharges are larger (median 64 X / 76 Y strips) and now span essentially the *whole* plane ($\sim$360 mm of 399 mm), versus half the plane in det3 — the higher gain lets a discharge propagate across the full active area. The Y/FEU8 plane again discharges harder than X, consistent with det7's known FEU8 saturation band. Only 14 % of spark hits saturate the ADC.

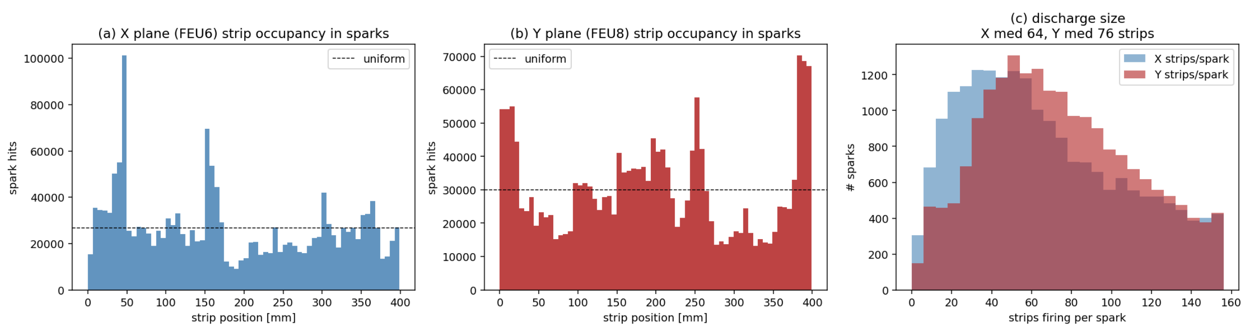

Figure 4: det7 locations: (a,b) edge peaks + hot strips in X and Y; (c) discharge size — larger than det3, Y harder than X.

The det7 spark hitmaps (Fig. 5) show the same fixed edge+hot-strip **grid** footprint as det3 (with an even more prominent X/FEU6 hot strip), over a central discharge centre — a pattern locked to detector geometry, not the muon flux, and so a shower veto in itself. The detector-side track check holds too: the widest contiguous fired-strip cluster is median **20 mm** (Fig. 6c), far wider than a $\sim$mm particle track, so the fired strips are discharge blobs, not tight shower tracks.

6 Q4 — Still no "walk", but longer-lived

det7 sparks are, like det3, neither simultaneous (0.8 % within 100 ns) nor propagating: the position–time correlation is $|r| \approx 0.15$ and the spark-centred Δpos–Δtime map (Fig. 7b) is again a cross with no tilt. The difference is duration: the median per-spark time spread is 334 ns (vs det3's 253 ns) and **66 %** of det7 sparks spread beyond 300 ns (vs 24 % for det3) — the discharge rings/afterpulses

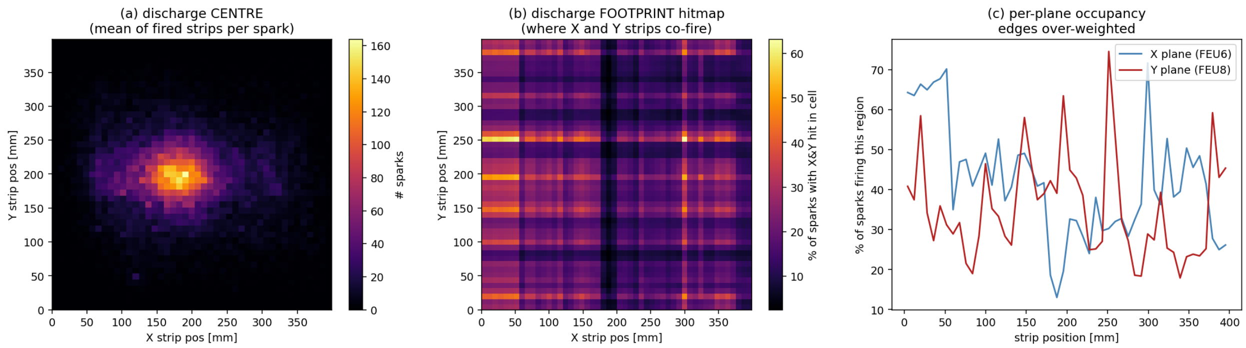

Figure 5: det7 spark hitmaps: (a) discharge centre; (b) footprint grid (edges + hot strips); (c) per-plane occupancy.

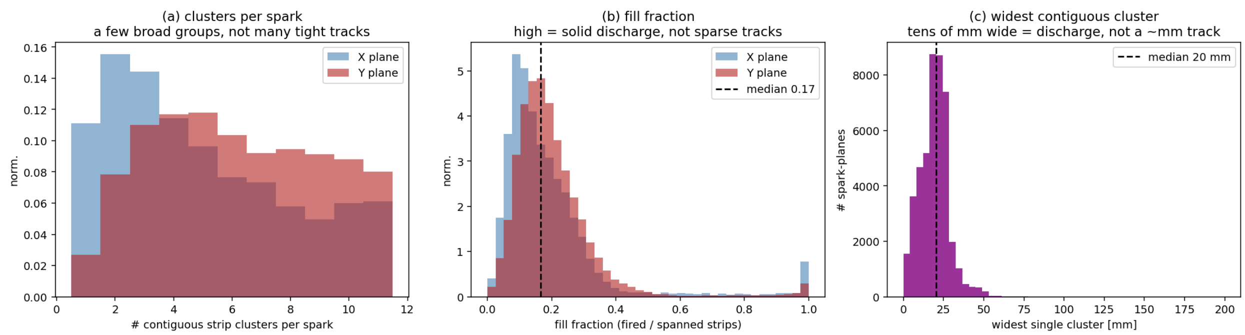

Figure 6: det7 detector-side shower check: (a) clusters/spark; (b) fill fraction; (c) widest contiguous cluster (median 20 mm $\gg$ a particle track).

for longer, filling much of the readout window (Fig. 7a,c). Same mechanism (extended discharge + afterpulsing, timing independent of position), longer tail.

7 Takeaways

- det7 and det3 spark by the **same mechanism**; det7 is simply operated further into breakdown, which raises the rate (0.33→1.61 Hz), the muon enhancement (4.1→5.9×), the spontaneous floor (1.2→5.3 %), the discharge size (half→whole plane) and its duration (253→334 ns).
- The lever is the **operating gain**: det7 needs a lower resist HV (the HV-scan optimum was ∼440 V; this run sat above it). Lowering it cuts the rate but not the muon-seeded nature.
- The **edge / hot-strip** weak spots and the **DAQ common-mode cross-talk** (spark → M3 track lost) reproduce on a second detector, so both are systemic, not det3 quirks — worth fixing at the detector-edge/guard-ring and DAQ level.

Reproduce: `extract_sparks.py g_det7_long` then `make_plots.py g_det7_long`; numbers in `spark_meta.json`. Identical pipeline to `det3_spark_analysis`.

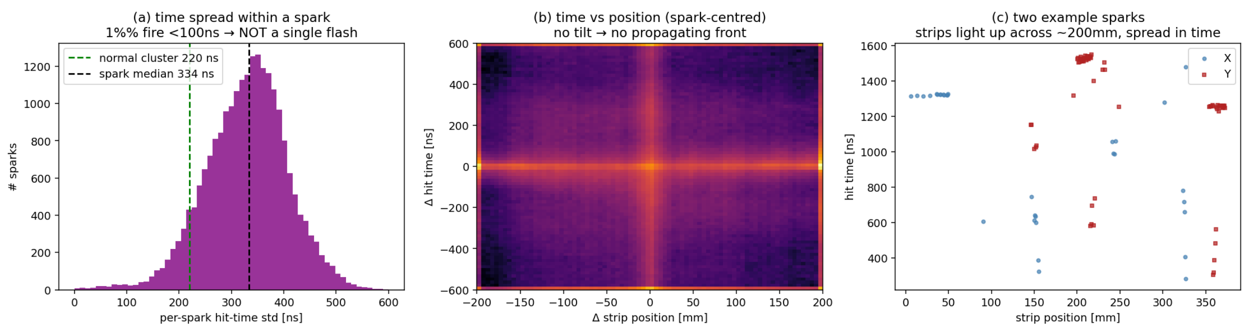

Figure 7: det7 time structure: (a) longer per-spark spread than det3; (b) no tilt (no propagating front); (c) example sparks across the whole plane, spread in time.